

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company discovers that its inventory management system cannot support newly introduced government tax regulations. Developers modify the software to comply with the updated regulations. Which type of software maintenance is required?

- A. Corrective Maintenance
- B. Adaptive Maintenance
- C. Perfective Maintenance
- D. Preventive Maintenance

Answer: B

2. Which of the following is considered the primary objective of Software Engineering?

- A. Increase programming complexity
- B. Develop high-quality software within time and budget
- C. Eliminate software testing
- D. Reduce hardware costs only

Answer: B

3. Which Software Development Life Cycle (SDLC) model is most appropriate when customer requirements are well understood but project risks are high?

- A. Waterfall
- B. Spiral
- C. Prototype
- D. Scrum

Answer: B

4. During software development, customer feedback is collected after each increment before the next increment begins. Which process model follows this approach?

- A. Incremental Model
- B. Waterfall Model
- C. V-Model
- D. Big Bang Model

Answer: A

5. Which document formally describes all software requirements before system design begins?

- A. Test Plan
- B. User Manual
- C. Software Requirements Specification (SRS)

D. Project Proposal

Answer: C

6. Which requirement specifies that an ATM system must dispense cash within five seconds after authentication?

- A. Business Requirement
- B. Functional Requirement
- C. User Requirement
- D. Non-functional Requirement

Answer: D

7. Which Requirements Engineering activity focuses on collecting information from customers and stakeholders?

- A. Validation
- B. Verification
- C. Elicitation
- D. Maintenance

Answer: C

8. Frequent uncontrolled changes requested by stakeholders after project approval are known as—

- A. Configuration Audit
- B. Scope Creep
- C. Refactoring
- D. Requirement Validation

Answer: B

9. Which Agile principle encourages welcoming requirement changes even late in development?

- A. Detailed Documentation
- B. Customer Collaboration
- C. Responding to Change
- D. Fixed Planning

Answer: C

10. Which Agile practice divides development into fixed-length iterations?

- A. Refactoring
- B. Timeboxing
- C. Pair Programming
- D. Code Inspection

Answer: B

11. In Extreme Programming (XP), who writes acceptance tests to verify customer requirements?

- A. Customer
- B. Tracker
- C. Coach
- D. Programmer

Answer: A

12. Which XP practice automatically combines newly developed code with the existing system several times a day?

- A. Pair Programming
- B. Refactoring
- C. Continuous Integration
- D. Coding Standards

Answer: C

13. Which Scrum artifact contains all desired features and enhancements for the entire product?

- A. Sprint Goal
- B. Sprint Backlog
- C. Product Backlog
- D. Burndown Chart

Answer: C

14. Which Scrum role is responsible for ensuring that Scrum principles are properly followed?

- A. Product Owner
- B. Scrum Master
- C. Client
- D. Business Analyst

Answer: B

15. According to the MoSCoW prioritization technique, which requirement should normally be implemented if time permits but is not absolutely essential?

- A. Must Have
- B. Should Have
- C. Could Have
- D. Won't Have

Answer: C

16. Which DSDM principle emphasizes frequent delivery of useful software rather than waiting until project completion?

- A. Reversible Changes
- B. Deliver on Time
- C. Demonstrate Control
- D. Focus on Business Need

Answer: B

17. Which is the first process in Feature Driven Development (FDD)?

- A. Build Feature List
- B. Design by Feature
- C. Develop Overall Model
- D. Plan by Feature

Answer: C

18. In Feature Driven Development, who is primarily responsible for coordinating feature implementation among developers?

- A. Project Manager
- B. Chief Programmer
- C. Customer
- D. Scrum Master

Answer: B

19. Which of the following best describes a software myth?

- A. Software quality depends on customer feedback.
- B. More developers always finish a late project faster.
- C. Testing improves software quality.
- D. Software maintenance is part of the SDLC.

Answer: B

20. Which activity verifies that software products satisfy specified requirements before delivery?

- A. Coding
- B. Verification and Validation
- C. Deployment
- D. Requirement Elicitation

Answer: B